

Network final - coding / EDA

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The AGORA (AI Governance and Regulatory Archive) dataset is a living collection of AI-relevant laws, regulations, standards, and other governance documents from the United States and around the world. AGORA includes document text, metadata, summaries, and thematic tags to enable deep analysis of the global AI governance landscape.

This section provides basic EDA for the main categories used in our network graph.

```
# imports
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
from collections import Counter
import nltk
from nltk.corpus import stopwords
from nltk.tokenize import word_tokenize
from nltk.tokenize import sent_tokenize
from sklearn.feature_extraction.text import CountVectorizer
import os as os
import shutil as shutil
import glob
from nltk.sentiment.vader import SentimentIntensityAnalyzer
df = pd.read_csv("documents.csv")
```

```
collection_counts = df['Collections'].value_counts()

print("total number of txt files per Collection type:")
print(collection_counts.to_string())
```

```
total number of txt files per Collection type:
Collections
```

U.S. federal laws	572
U.S. state and local documents	345
U.S. regulations, executive orders, and agency policies	37
NDAA provisions;U.S. federal laws;FY2026 NDAA	37
U.S. federal laws;NDAA provisions;FY2023 NDAA	30
Chinese law and policy	26
Miscellaneous documents	25
U.S. federal laws;NDAA provisions;FY2021 NDAA	24
NDAA provisions;FY2025 NDAA	23
U.S. federal laws;NDAA provisions;FY2024 NDAA	22
Corporate policies and commitments	12
U.S. federal laws;NDAA provisions;FY2022 NDAA	12
Editors' Picks;U.S. regulations, executive orders, and agency policies	10
Multinational	10
Editors' Picks;U.S. state and local documents	9
Editors' Picks;Multinational	9
Editors' Picks;Chinese law and policy	7
U.S. regulations, executive orders, and agency policies;Editors' Picks	6
Editors' Picks;Corporate policies and commitments	5
Editors' Picks;Miscellaneous documents	4
Editors' Picks	4
Editors' Picks;U.S. federal laws	1
U.S. state and local documents;Editors' Picks	1
Chinese law and policy;Editors' Picks	1

The code above looks at the documents.csv file to determine the number of .txt files that are included in each Collection. It looks like some topics have mutiple subcategories for Collections (ex. U.S. federal laws;NDAA provisions; FY2023 NDAA and U.S. federal laws; NDAA provisions;FY2022 NDAA) while others are more generalized.

The cell below filters the documents.csv file to provide a list of matching Agora IDs. these IDs correspond to the .txt file that contains the contents of that policy.

```
# filtering txt files by collection

def get_agora_ids_by_collection(df, collections):
    """
    filters a .csv file by the given collection and returns a list of matching Agora IDs
    """

    # clean whitespace
    matched_rows = df[df['Collections'].astype(str).str.strip() == collections.strip()]
```

```

# extract ids, drop nulls, convert to str
id_list = matched_rows['AGORA ID'].dropna().astype(str).tolist()

return id_list

```

The cell below uses the `get_agora_ids_by_collection` function to match IDs with the corresponding .txt file, filtering by collection. This scope of this EDA section is limited to U.S. regulations, executive orders, and agency policies, U.S. state and local documents, and U.S. federal laws.

```

# extracting targeted collection
target_collection = "Multinational"
matching_ids = get_agora_ids_by_collection(df, target_collection)
# id_string = ", ".join(str(id) for id in matching_ids)

# view results
# print(f"IDs found for '{target_collection}':", matching_ids)
# print(f'total number of matches: {len(matching_ids)}')

```

The cell below evaluates the .txt files in the fulltext folder to match them to the collection type, and then creates a new folder for .txt files that are included in that collection.

```

def group_collection_text_files(df, collection_name, source_folder="fulltext", output_base_f
    """
    looks inside the 'fulltext' folder for {id}.txt files matching a collection and copies th
    """

    # filter and extract integer IDs, drop NA's, make int, convert to list
    matched_rows = df[df['Collections'].astype(str).str.strip() == collection_name.strip()]
    matching_ids = matched_rows['AGORA ID'].dropna().astype(int).tolist()

    print(f"found {len(matching_ids)} matching records in the documents csv for: '{collection

    if not matching_ids:
        print("no matching IDs found for this collection name.")
        return matching_ids

    # 3. create destination folder
    safe_folder_name = "".join(c for c in collection_name if c.isalnum() or c in (' ', '_'),
    destination_folder = os.path.join(output_base_folder, safe_folder_name)

```

```

os.makedirs(destination_folder, exist_ok=True)
print(f"created/verified destination folder at: '{destination_folder}'")

# 4. search 'fulltext' folder and copy corresponding '{id}.txt' files
copied_count = 0
for agora_id in matching_ids:
    filename = f"{agora_id}.txt"

    # look INSIDE the fulltext subfolder
    source_file_path = os.path.join(source_folder, filename)
    # put INSIDE the newly created collection folder
    destination_file_path = os.path.join(destination_folder, filename)

    if os.path.exists(source_file_path):
        shutil.copy2(source_file_path, destination_file_path)
        copied_count += 1

print(f"successfully copied {copied_count} out of {len(matching_ids)} text files into '{destination_folder}'")
return matching_ids

```

EDA

The cell below applies the `group_collection_text_files` function to the targeted collections for this EDA section. This was repeated for each collection to create new subfolders for them.

```

# running function
target_collection = "Corporate policies and commitments"
matching_ids = group_collection_text_files(df, target_collection)

```

```

found 12 matching records in the documents csv for: 'Corporate policies and commitments'
created/verified destination folder at: './Corporate policies and commitments'
successfully copied 12 out of 12 text files into 'Corporate policies and commitments/'

```

The cell below reads all the .txt files in the specified collection folder and provides information on word count, file length, and word frequencies.

```

def eda_function(folder_path, stop_words_lang = 'english'):
    """
    reads all txt files in the collection folders and performs text EDA focusing
    on word counts, file length, and word frequencies
    """

```

```

search_path = os.path.join(folder_path, '*.txt')
file_paths = glob.glob(search_path)

print(f"text EDA report for folder: '{folder_path}'")
print(f"total text files found: {len(file_paths)}")

# metrics
data_metrics = []
all_words_compiled = []
stop_words_set = set(stopwords.words(stop_words_lang))

for path in file_paths:
    file_name = os.path.basename(path)
    # print(f"DEBUG: attempting to read file -> {file_name}")

    try:
        with open(path, 'r', errors = 'ignore') as f:
            raw_text = f.read().strip()
    except Exception as e:
        print(f"read error on '{file_name}': {e}")
        continue
    if not raw_text:
        print(f"SKIPPED: '{file_name}' is completely empty (0 characters).")
        continue

    # convert text structure
    char_counts = len(raw_text)
    words = word_tokenize(raw_text)
    sentences = sent_tokenize(raw_text)

    word_count = len(words)
    sentence_count = len(sentences)

    # avg word length
    avg_word_length = np.mean([len(w) for w in words]) if word_count > 0 else 0

    # frequency analysis
    clean_file_words = [w.lower() for w in words if w.isalpha()]
    all_words_compiled.extend(clean_file_words)

    # entry metrics
    data_metrics.append({

```

```

        'file name' : file_name,
        'charecter length': char_counts,
        'word count': word_count,
        'sentence count': sentence_count,
        'avg word length': avg_word_length,
    })
    # print(f"parsed '{file_name}' successfully")

metrics_df = pd.DataFrame(data_metrics)
if metrics_df.empty:
    print('warning: no readable data extracted from .txt files')
    return None

    # statistical summaries
print('length and summary profiles:')
print(metrics_df[['charecter length', 'word count', 'sentence count', 'avg word length']])

    # frequency
meaningful_words = [w for w in all_words_compiled if w not in stop_words_set]
word_frequencies = Counter(meaningful_words)

print('top 10 most frequent meaningful words, excluding stop words')
for word, count in word_frequencies.most_common(10):
    print(f"    • {word}: {count}")

    # visualizations
fig, axes = plt.subplots(1,2, figsize = (10, 6))

    # word count distributions
sns.histplot(metrics_df['word count'], kde = True, ax = axes[0], color = 'blue', bins = 10)
axes[0].set_title('distribution of file word counts')
axes[0].set_xlabel('words per file')

    # file length
sns.histplot(metrics_df['charecter length'], kde = True, ax=axes[1], color='indigo', bins = 10)
axes[1].set_title('distribution of file character lengths')
axes[1].set_xlabel('characters per file')

plt.tight_layout()
plt.show()

```

```
return metrics_df
```

Corporate policies and commitments - EDA

```
folder_to_analyze = "Corporate policies and commitments"  
analysis_results_df = eda_function(folder_path=folder_to_analyze)
```

text EDA report for folder: 'Corporate policies and commitments'

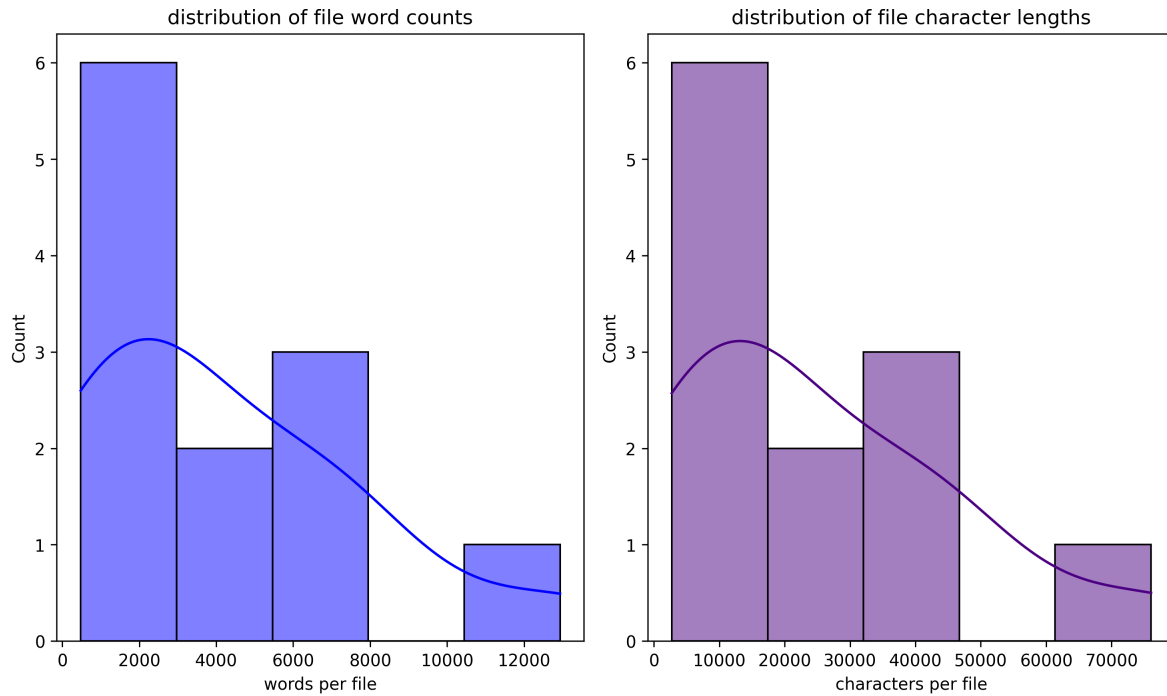
total text files found: 12

length and summary profiles:

	charecter length	word count	sentence count	avg word length
count	12.000000	12.000000	12.000000	12.000000
mean	25199.333333	4245.000000	143.833333	5.026834
std	22045.415270	3724.178421	132.960577	0.104457
min	2732.000000	481.000000	14.000000	4.804574
25%	7412.250000	1249.250000	44.500000	4.991053
50%	17832.500000	2997.000000	88.000000	5.039932
75%	42305.750000	7031.000000	208.750000	5.079848
max	76039.000000	12935.000000	437.000000	5.193269

top 10 most frequent meaningful words, excluding stop words

- ai: 756
- model: 268
- models: 255
- system: 206
- use: 203
- safety: 199
- data: 199
- security: 197
- systems: 187
- risks: 174



Multinational - EDA

```
folder_to_analyze = "Multinational"
analysis_results_df = eda_function(folder_path=folder_to_analyze)
```

text EDA report for folder: 'Multinational'

total text files found: 10

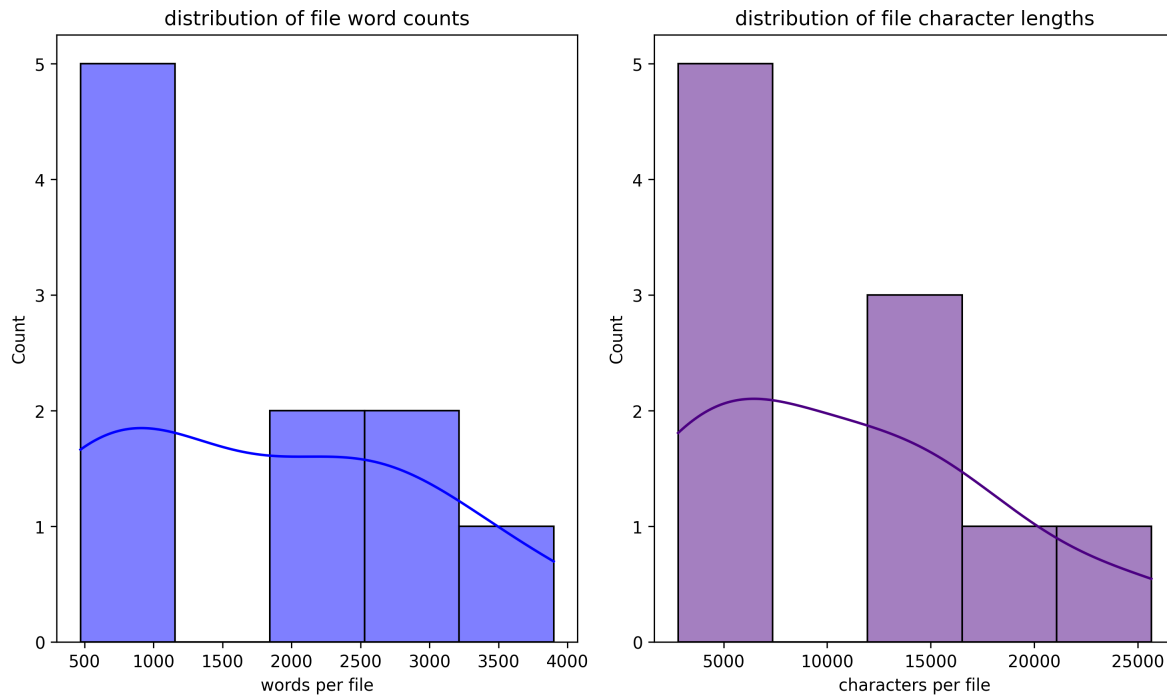
length and summary profiles:

	charecter length	word count	sentence count	avg word length
count	10.000000	10.000000	10.000000	10.000000
mean	10765.500000	1786.400000	40.200000	5.137822
std	7478.468281	1191.789989	25.823331	0.388437
min	2800.000000	470.000000	14.000000	4.230506
25%	4839.250000	758.000000	23.000000	5.065992
50%	10059.000000	1699.500000	40.000000	5.138927
75%	13610.250000	2556.500000	44.000000	5.303811
max	25650.000000	3900.000000	104.000000	5.673077

top 10 most frequent meaningful words, excluding stop words

- ai: 363
- systems: 147
- development: 111
- intelligence: 108

- artificial: 107
- system: 93
- risk: 87
- including: 81
- international: 78
- human: 67



US state and local documents - EDA

```
folder_to_analyze = "US state and local documents"
analysis_results_df = eda_function(folder_path=folder_to_analyze)
```

text EDA report for folder: 'US state and local documents'

total text files found: 345

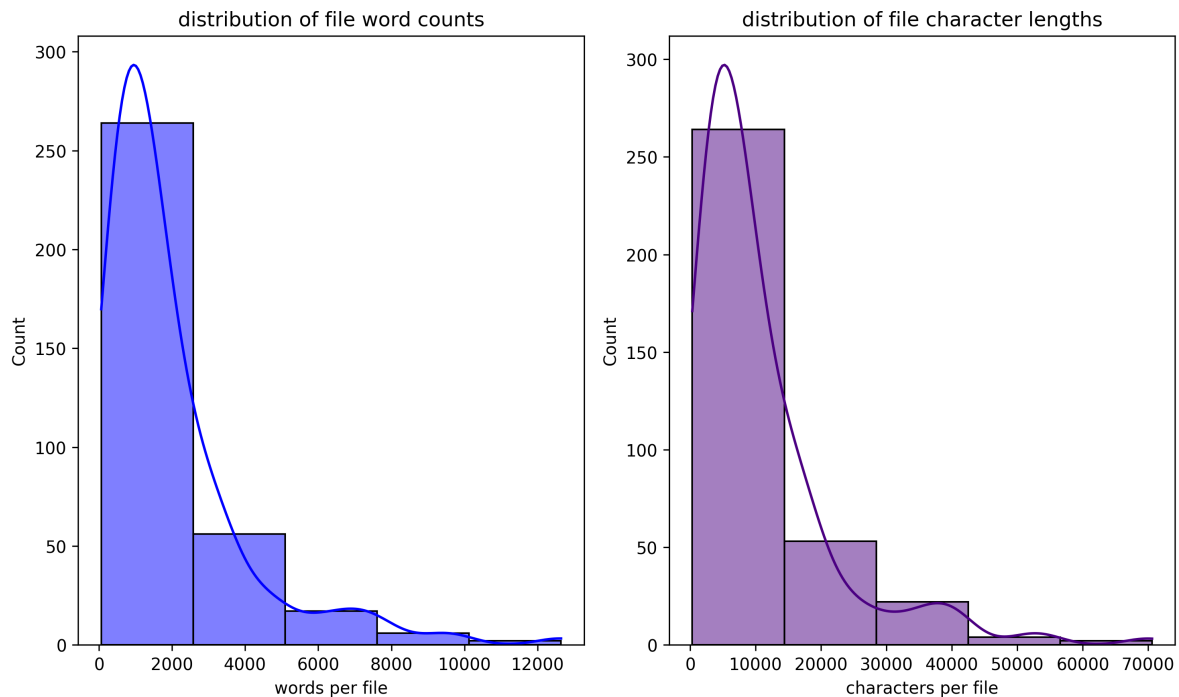
length and summary profiles:

	charecter length	word count	sentence count	avg word length
count	345.000000	345.000000	345.000000	345.000000
mean	10754.556522	1950.394203	55.860870	4.632397
std	10984.695933	2003.238873	56.724685	0.297351
min	329.000000	59.000000	1.000000	3.795673
25%	3768.000000	700.000000	19.000000	4.446234
50%	7205.000000	1301.000000	39.000000	4.615164

75%	13192.000000	2446.000000	70.000000	4.797587
max	70618.000000	12646.000000	449.000000	5.737485

top 10 most frequent meaningful words, excluding stop words

- section: 4689
- shall: 4673
- intelligence: 3063
- artificial: 3057
- state: 2649
- ai: 2552
- information: 2463
- use: 2431
- b: 2319
- system: 2206



the top 10 most frequent meaningful words excluding stop words for the “US state and local documents” folder were:

- section: 4689 • shall: 4673 • intelligence: 3063 • artificial: 3057 • state: 2649 • ai: 2552
- information: 2463 • use: 2431 • b: 2319 • system: 2206

These documents are typically under 2000 words, which is on average shorter than some of the other texts used in our network.

US regulations executive orders and agency policies - EDA

```
folder_to_analyze = "US regulations executive orders and agency policies"
analysis_results_df = eda_function(folder_path=folder_to_analyze)
```

text EDA report for folder: 'US regulations executive orders and agency policies'

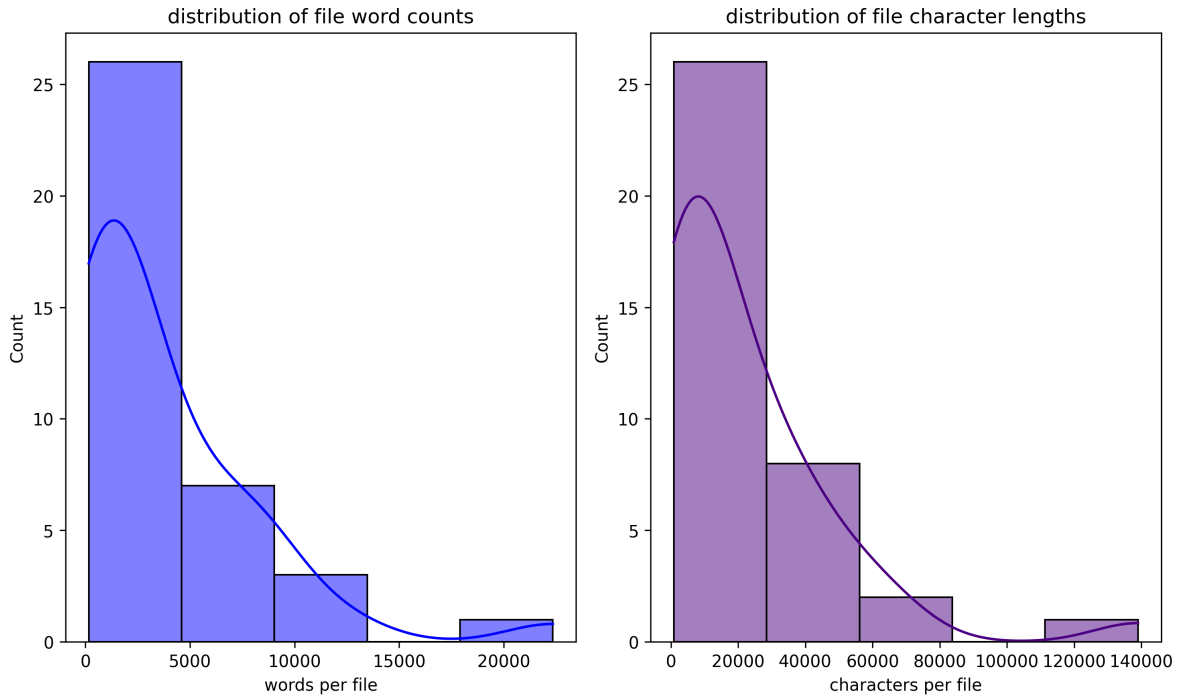
total text files found: 37

length and summary profiles:

	charecter length	word count	sentence count	avg word length
count	37.000000	37.000000	37.000000	37.000000
mean	21206.621622	3701.837838	101.216216	4.831365
std	27021.611165	4537.846403	123.368269	0.314888
min	874.000000	163.000000	7.000000	3.735101
25%	3446.000000	610.000000	21.000000	4.612097
50%	10496.000000	1773.000000	51.000000	4.860370
75%	30116.000000	5064.000000	135.000000	5.053120
max	138992.000000	22344.000000	608.000000	5.336153

top 10 most frequent meaningful words, excluding stop words

- ai: 1762
- use: 550
- data: 519
- agencies: 444
- must: 431
- information: 428
- systems: 420
- may: 413
- including: 389
- federal: 303



These documents are typically under 5000 words. The top 10 most frequent meaningful words give us some insight into the focus for AI at the federal level, although further analysis is needed.

The National Defense Authorization Act is any of a series of United States federal laws specifying the annual budget and expenditures of the U.S. Department of Defense. The dataset provides .txt data for 2021-2024 and 2026

US federal lawsNDAA provisionsFY2021 NDAA - EDA

```
folder_to_analyze = "US federal lawsNDAA provisionsFY2021 NDAA"
analysis_results_df = eda_function(folder_path=folder_to_analyze)
```

text EDA report for folder: 'US federal lawsNDAA provisionsFY2021 NDAA'

total text files found: 24

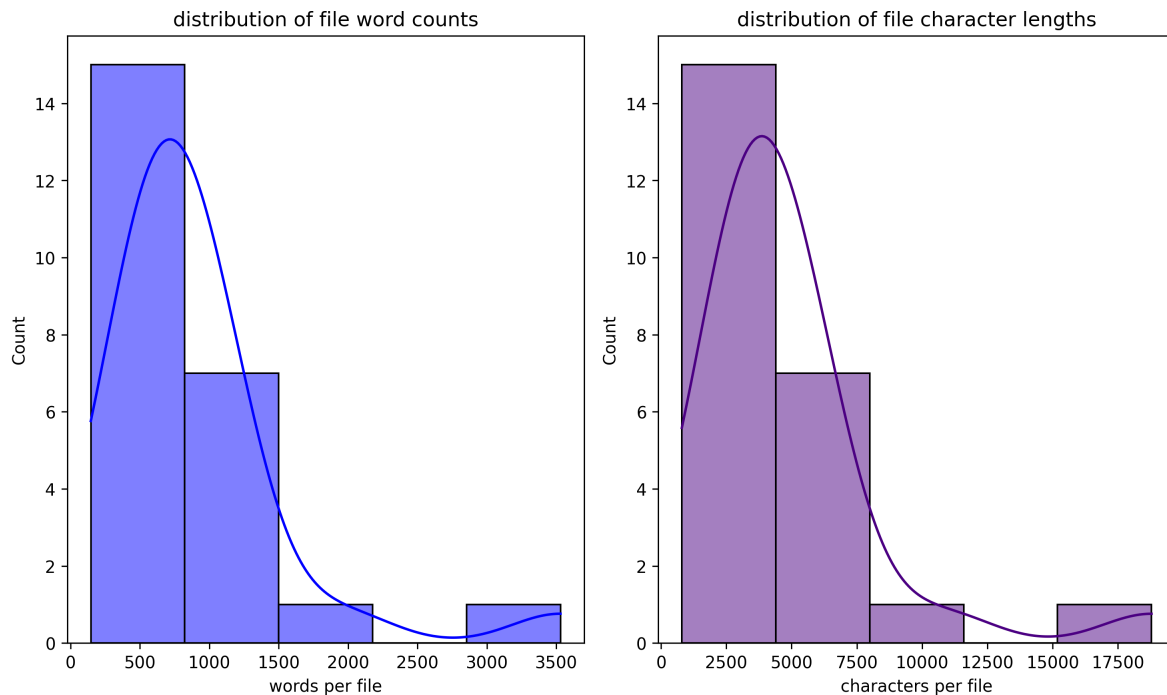
length and summary profiles:

	charecter length	word count	sentence count	avg word length
count	24.000000	24.000000	24.000000	24.000000
mean	4858.416667	900.375000	18.708333	4.608577
std	3569.336047	670.423799	13.668368	0.249451
min	796.000000	146.000000	6.000000	4.011583
25%	3010.000000	560.250000	9.000000	4.492789
50%	3747.500000	693.500000	16.500000	4.625785

75%	5557.750000	1046.250000	22.250000	4.788770
max	18784.000000	3531.000000	68.000000	5.141066

top 10 most frequent meaningful words, excluding stop words

- defense: 190
- secretary: 166
- shall: 162
- b: 128
- department: 106
- section: 105
- intelligence: 96
- artificial: 90
- including: 89
- subsection: 87



US federal lawsNDAA provisionsFY2022 NDAA - EDA

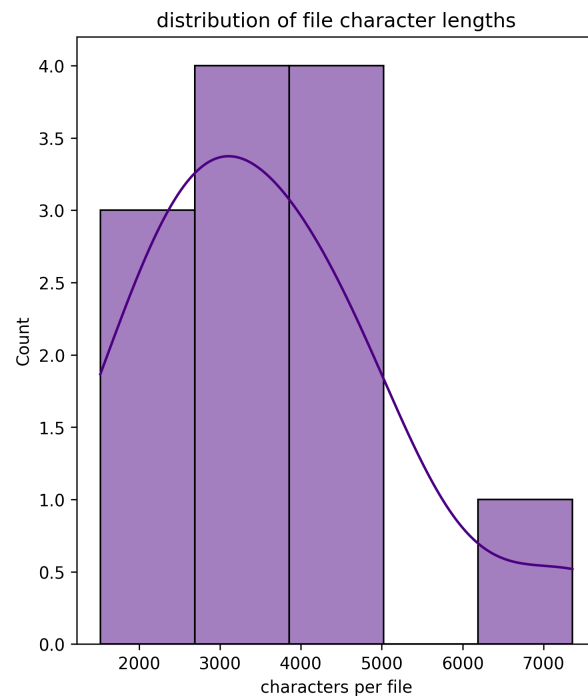
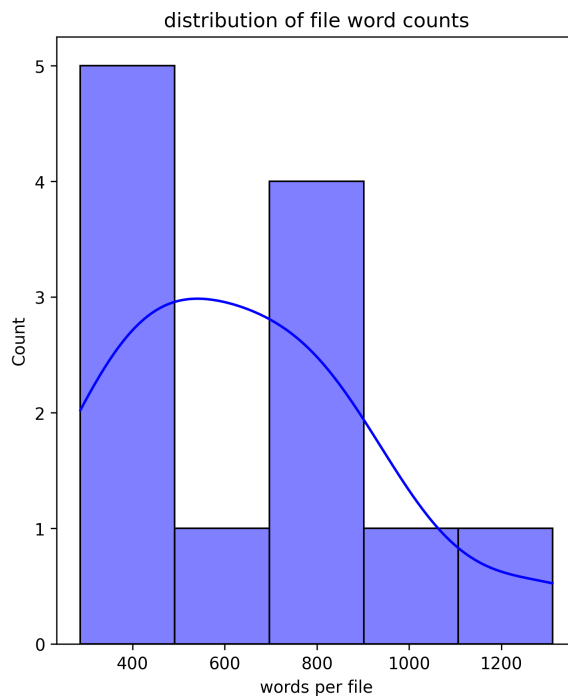
```
folder_to_analyze = "US federal lawsNDAA provisionsFY2022 NDAA"
analysis_results_df = eda_function(folder_path=folder_to_analyze)
```

text EDA report for folder: 'US federal lawsNDAA provisionsFY2022 NDAA'
total text files found: 12
length and summary profiles:

	charecter length	word count	sentence count	avg word length
count	12.000000	12.000000	12.000000	12.000000
mean	3597.666667	650.833333	15.583333	4.734543
std	1585.584339	291.494841	9.976867	0.364891
min	1522.000000	287.000000	6.000000	3.924242
25%	2689.250000	434.000000	9.750000	4.622303
50%	3229.500000	626.000000	12.000000	4.774326
75%	4412.250000	789.000000	19.500000	4.858388
max	7356.000000	1311.000000	38.000000	5.508046

top 10 most frequent meaningful words, excluding stop words

- defense: 106
- secretary: 86
- intelligence: 63
- shall: 57
- department: 57
- artificial: 44
- b: 42
- subsection: 38
- data: 33
- capabilities: 28



US federal lawsNDAA provisionsFY2023 NDAA - EDA

```
folder_to_analyze = "US federal lawsNDAA provisionsFY2023 NDAA"
analysis_results_df = eda_function(folder_path=folder_to_analyze)
```

text EDA report for folder: 'US federal lawsNDAA provisionsFY2023 NDAA'

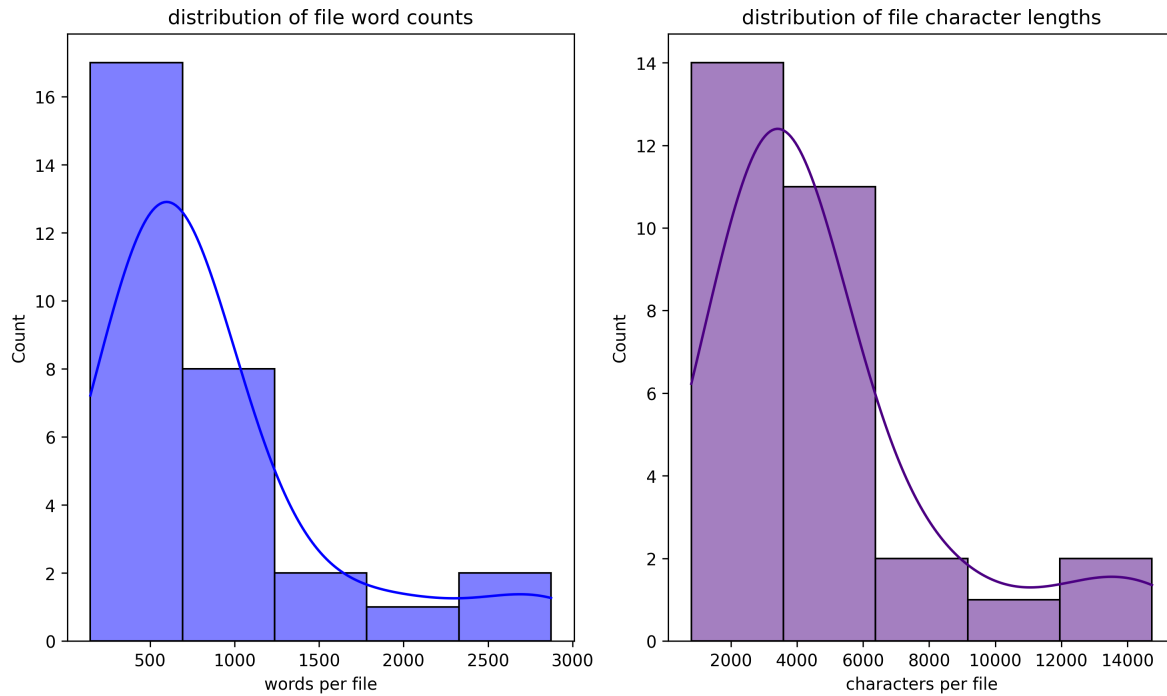
total text files found: 30

length and summary profiles:

	charecter length	word count	sentence count	avg word length
count	30.000000	30.000000	30.000000	30.000000
mean	4798.366667	870.133333	18.266667	4.787119
std	3473.479310	677.529016	13.513552	0.323062
min	805.000000	146.000000	4.000000	4.064045
25%	2775.500000	482.250000	8.250000	4.625168
50%	3668.000000	644.000000	13.500000	4.736046
75%	5158.250000	920.250000	21.000000	4.974188
max	14738.000000	2873.000000	55.000000	5.516279

top 10 most frequent meaningful words, excluding stop words

- intelligence: 272
- shall: 191
- b: 169
- section: 152
- artificial: 150
- defense: 142
- subsection: 114
- secretary: 102
- c: 95
- national: 91



US federal lawsNDAA provisionsFY2024 NDAA - EDA

```
folder_to_analyze = "US federal lawsNDAA provisionsFY2024 NDAA"
analysis_results_df = eda_function(folder_path=folder_to_analyze)
```

text EDA report for folder: 'US federal lawsNDAA provisionsFY2024 NDAA'

total text files found: 22

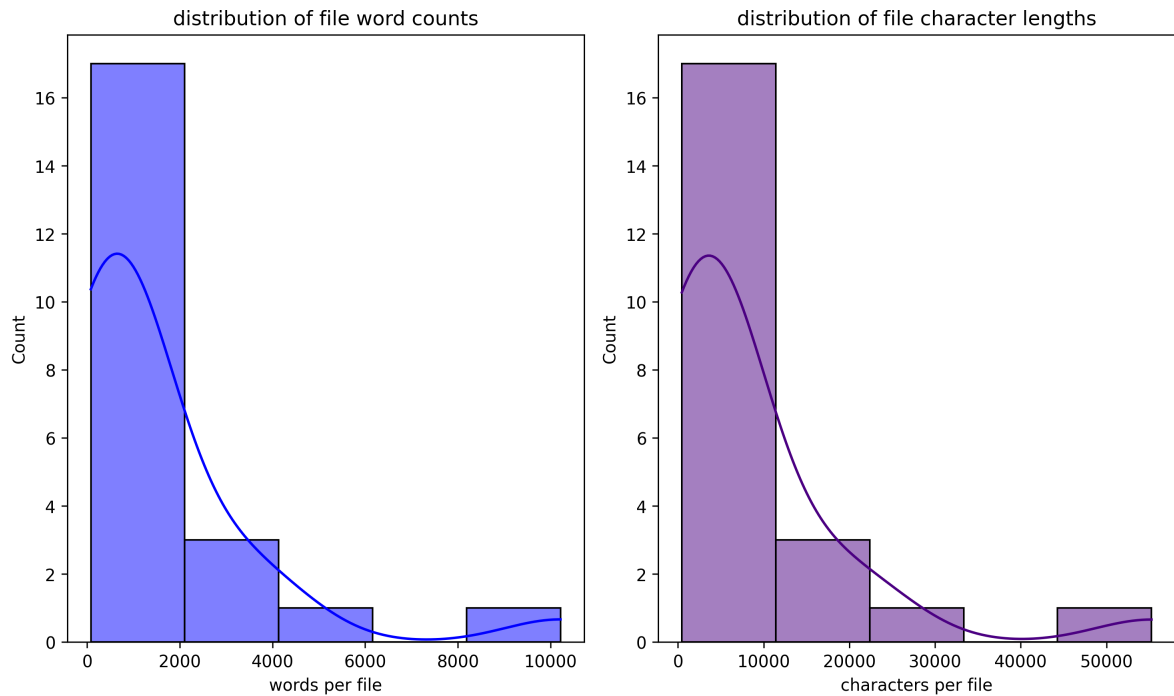
length and summary profiles:

	charecter length	word count	sentence count	avg word length
count	22.000000	22.000000	22.000000	22.000000
mean	8325.590909	1509.318182	34.227273	4.751270
std	12268.348015	2254.447993	48.319307	0.286571
min	453.000000	80.000000	4.000000	4.242647
25%	2508.500000	415.250000	7.500000	4.564761
50%	3472.000000	632.500000	11.000000	4.777406
75%	6137.000000	1130.250000	40.500000	4.985781
max	55253.000000	10217.000000	196.000000	5.286747

top 10 most frequent meaningful words, excluding stop words

- defense: 307
- shall: 266
- intelligence: 228
- united: 180

- department: 171
- artificial: 167
- subsection: 165
- b: 163
- secretary: 162
- section: 138



NDAA provisionsUS federal lawsFY2026 NDAA

```
folder_to_analyze = "NDAA provisionsUS federal lawsFY2026 NDAA"
analysis_results_df = eda_function(folder_path=folder_to_analyze)
```

text EDA report for folder: 'NDAA provisionsUS federal lawsFY2026 NDAA'

total text files found: 37

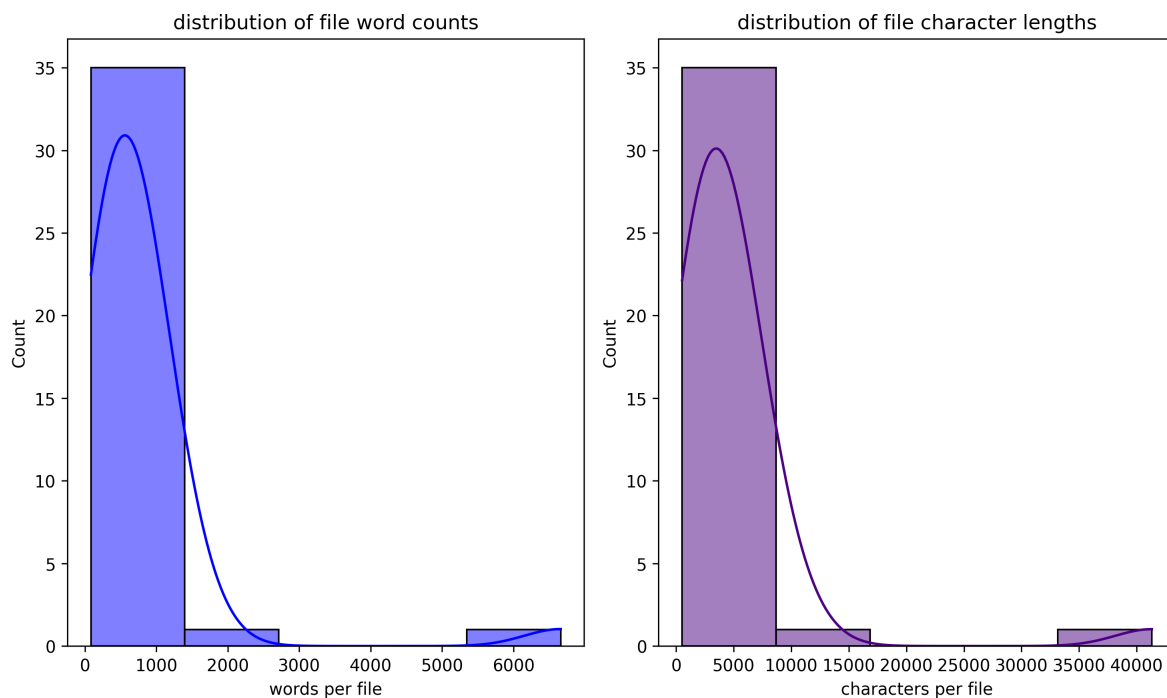
length and summary profiles:

	charecter length	word count	sentence count	avg word length
count	37.000000	37.000000	37.000000	37.000000
mean	4865.459459	769.864865	17.027027	4.823803
std	6560.185878	1050.619927	18.005380	0.271125
min	524.000000	83.000000	4.000000	4.060790
25%	2053.000000	344.000000	9.000000	4.657682
50%	3209.000000	523.000000	12.000000	4.811448

75%	5616.000000	900.000000	17.000000	5.042553
max	41323.000000	6661.000000	111.000000	5.416490

top 10 most frequent meaningful words, excluding stop words

- intelligence: 295
- artificial: 245
- secretary: 230
- defense: 221
- shall: 208
- b: 160
- security: 140
- department: 128
- subsection: 127
- national: 118



Chinese law and policy - EDA

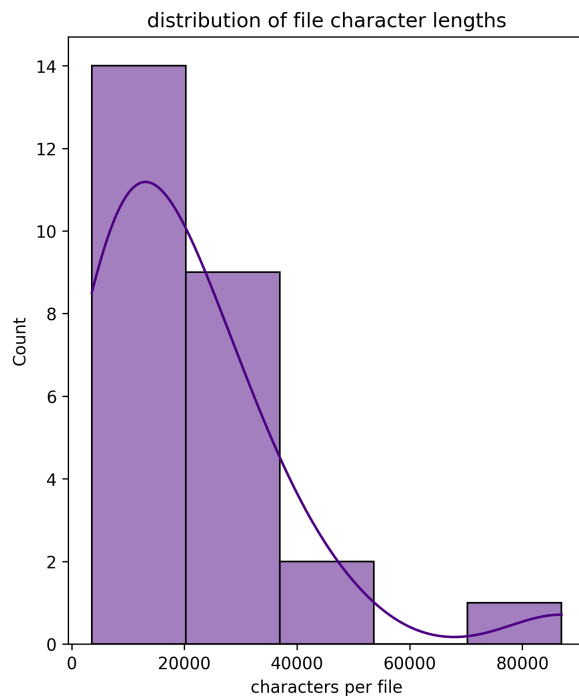
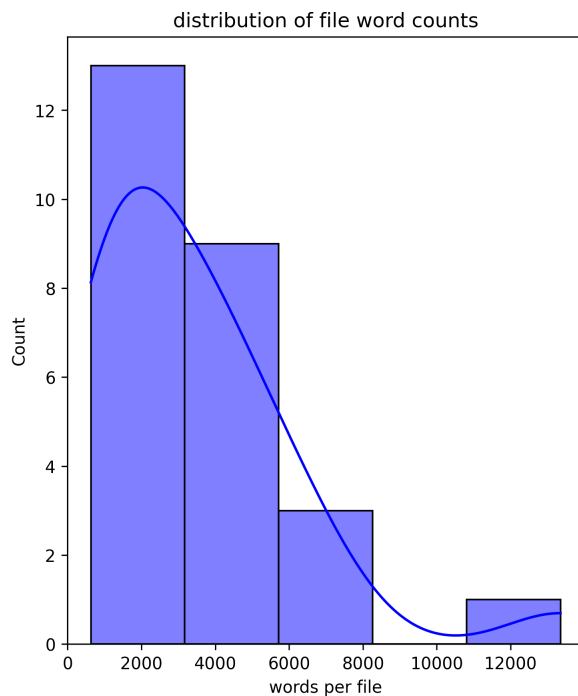
```
folder_to_analyze = "Chinese law and policy"
analysis_results_df = eda_function(folder_path=folder_to_analyze)
```

```
text EDA report for folder: 'Chinese law and policy'
total text files found: 26
length and summary profiles:
```

	charecter length	word count	sentence count	avg word length
count	26.000000	26.000000	26.000000	26.000000
mean	20580.269231	3366.500000	98.269231	5.216675
std	17962.657569	2805.063839	96.155107	0.371976
min	3563.000000	631.000000	12.000000	4.665917
25%	8639.000000	1369.000000	38.000000	4.910206
50%	16178.500000	2836.500000	91.500000	5.275794
75%	25353.250000	4427.000000	121.750000	5.482271
max	86969.000000	13351.000000	505.000000	5.945324

top 10 most frequent meaningful words, excluding stop words

- ai: 1269
- shall: 994
- ethics: 614
- development: 515
- data: 500
- technology: 465
- security: 437
- standards: 392
- research: 329
- relevant: 325



US Federal Laws - EDA

```
folder_to_analyze = "US federal laws"
analysis_results_df = eda_function(folder_path=folder_to_analyze)
```

text EDA report for folder: 'US federal laws'

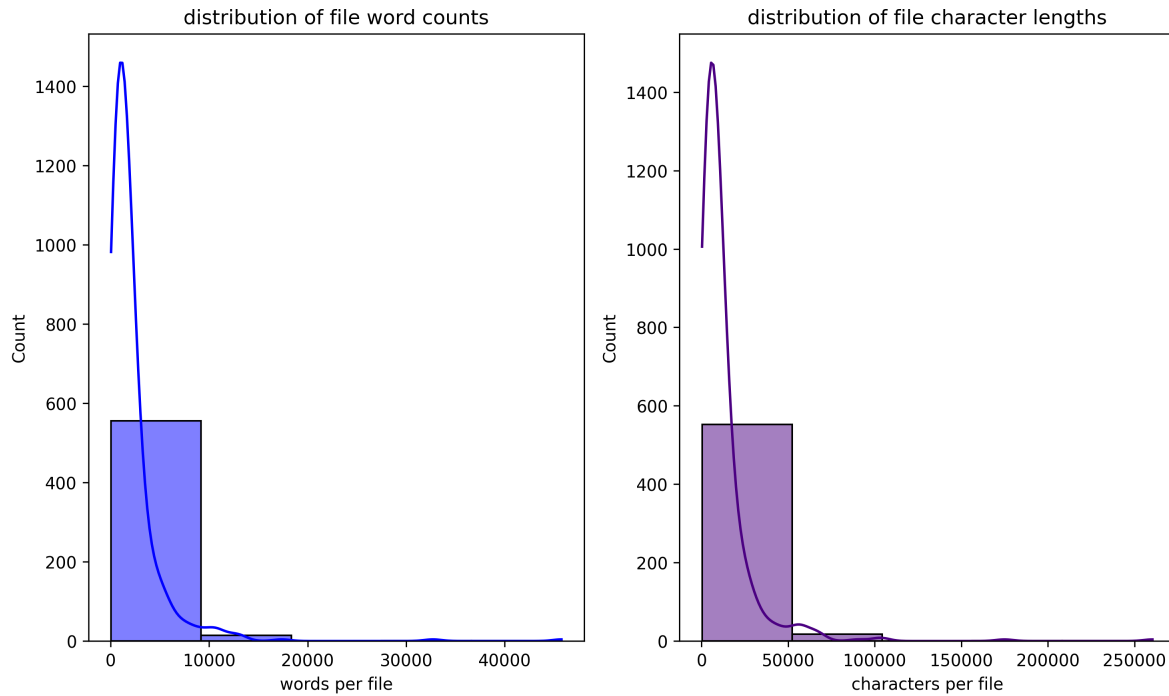
total text files found: 572

length and summary profiles:

	charecter length	word count	sentence count	avg word length
count	572.000000	572.000000	572.000000	572.000000
mean	11969.681818	2121.765734	45.975524	4.590590
std	18156.730025	3155.585289	63.346954	0.339679
min	353.000000	53.000000	1.000000	3.592000
25%	3785.500000	673.250000	14.000000	4.375349
50%	7490.000000	1316.500000	30.000000	4.585485
75%	12798.000000	2320.250000	51.000000	4.850612
max	260431.000000	45751.000000	866.000000	5.773585

top 10 most frequent meaningful words, excluding stop words

- shall: 7768
- b: 6482
- section: 5877
- act: 5633
- intelligence: 4986
- artificial: 4837
- including: 4302
- secretary: 3982
- states: 3844
- united: 3801



Sentiment Analysis

The function below calculates VADER sentiment score the the .txt files in each folder, and prints out a summary.

```
# sentiment analysis function

def sentiment_analyzer(folder_path):
    """
    reads all .txt files in a specific folder, calculates VADER sentiment score
    for each file, prints summary
    """
    search_path = os.path.join(folder_path, '*.txt')
    file_paths = glob.glob(search_path)

    print(f"sentiment analysis for folder: '{folder_path}")
    print(f'total text files found: {len(file_paths)}')

    if not file_paths:
        print('no .txt files found in this directory')
        return None

# initialize
```

```

sia = SentimentIntensityAnalyzer()
sentiment_data = []

for path in file_paths:
    file_name = os.path.basename(path)

    try:
        with open(path, 'r', errors = 'ignore') as f:
            raw_text = f.read().strip()
    except Exception as e:
        print(f"read error on '{file_name}': {e} ")
        continue
    if not raw_text:
        continue

    # VADAR scores
    scores = sia.polarity_scores(raw_text)

    # classifier

    if scores['compound'] >= 0.05:
        stance = 'Positive'
    elif scores['compound'] <= -0.05:
        stance = 'Negative'
    else:
        stance = 'Neutral'

    sentiment_data.append({
        'file name': file_name,
        'negative score': scores['neg'],
        'neutral score': scores['neu'],
        'positive score': scores['pos'],
        'compound score': scores['compound'],
        'overall stance': stance })

# make df
sentiment_df = pd.DataFrame(sentiment_data)

# show summary data

# numerical summaries
print('sentiment intensity metrics summary')

```

```

print(sentiment_df[['negative score', 'neutral score', 'positive score', 'compound score'])

# categorical distributions
print('categorical distributions ')
print(sentiment_df['overall stance'].value_counts().to_string())

# 3. graphs
fig, axes = plt.subplots(1, 2, figsize=(14, 5))

# compound score
sns.histplot(sentiment_df['compound score'], kde=True, ax=axes[0], color='blue', bins=10)
axes[0].set_title('distribution of compound sentiment scores')
axes[0].set_xlabel('compound score (-1 = negative to +1 = positive)')

# categorical breakdown
sns.countplot(data=sentiment_df, x='overall stance', hue = 'overall stance' , ax=axes[1])
axes[1].set_title('document stance classification counts')
axes[1].set_xlabel('stance category')
axes[1].set_ylabel('number of files')

plt.tight_layout()
plt.show()

return sentiment_df

```

Corporate policies and commitments - Sentiment Analysis

```

target_folder = "Corporate policies and commitments"

sentiment_results_df = sentiment_analyzer(folder_path=target_folder)

```

```

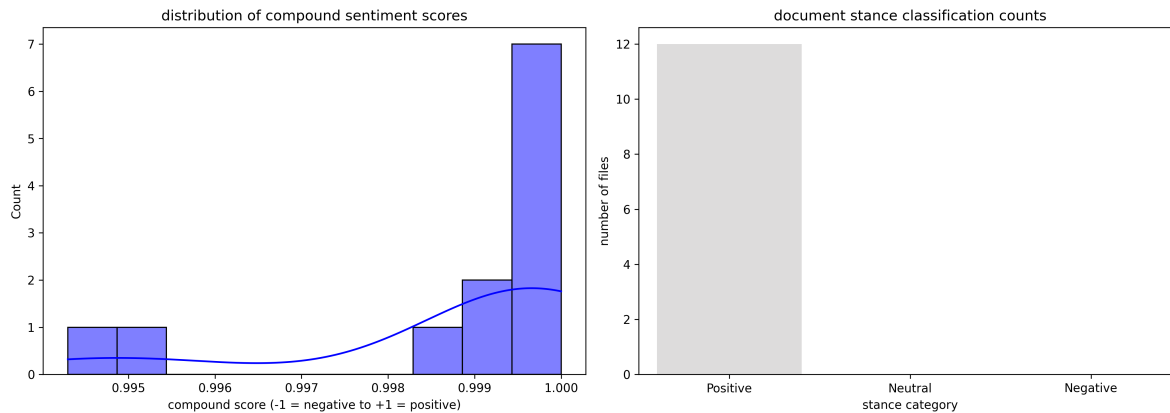
sentiment analysis for folder: 'Corporate policies and commitments'
total text files found: 12
sentiment intensity metrics summary

```

	negative score	neutral score	positive score	compound score
count	12.000000	12.000000	12.000000	12.000000
mean	0.052167	0.790833	0.157250	0.998833
std	0.018727	0.052641	0.046838	0.001919
min	0.024000	0.717000	0.055000	0.994300
25%	0.039000	0.772750	0.138750	0.999000
50%	0.052500	0.780500	0.158000	0.999700

75%	0.067750	0.811500	0.178750	0.999925
max	0.079000	0.920000	0.242000	1.000000

categorical distributions
overall stance
Positive 12



Sentiment from these documents are reported as positive.
Mutlinational — Sentiment Analysis

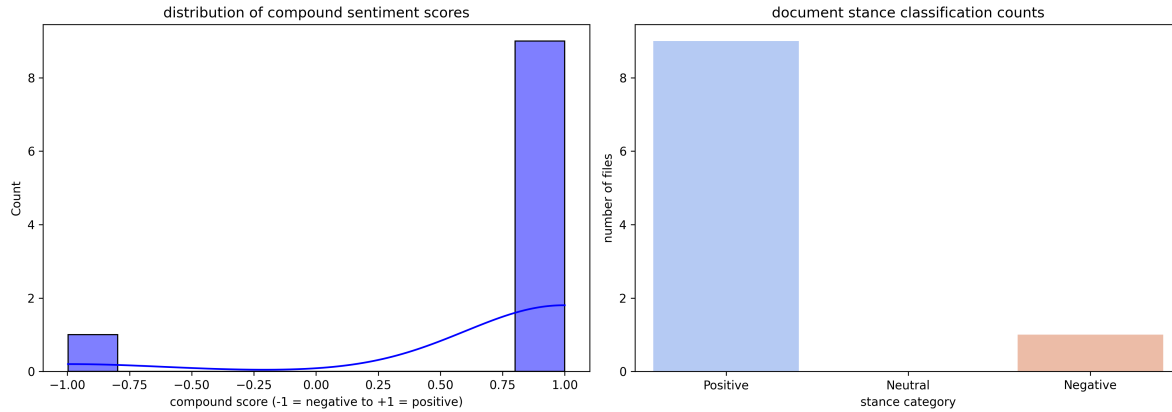
```
target_folder = "Multinational"

sentiment_results_df = sentiment_analyzer(folder_path=target_folder)
```

```
sentiment analysis for folder: 'Multinational'
total text files found: 10
sentiment intensity metrics summary
```

	negative score	neutral score	positive score	compound score
count	10.000000	10.000000	10.000000	10.000000
mean	0.053100	0.782000	0.165000	0.796370
std	0.036134	0.060579	0.060531	0.630281
min	0.009000	0.724000	0.044000	-0.997300
25%	0.018000	0.733250	0.134250	0.995050
50%	0.056000	0.763000	0.187000	0.998450
75%	0.077750	0.807750	0.201000	0.999650
max	0.109000	0.888000	0.246000	1.000000

categorical distributions
overall stance
Positive 9
Negative 1



Mostly positive with one document for negative sentiment.

US regulations executive orders and agency policies - Sentiment Analysis

```
target_folder = "US regulations executive orders and agency policies"
sentiment_results_df = sentiment_analyzer(folder_path=target_folder)
```

sentiment analysis for folder: 'US regulations executive orders and agency policies'
total text files found: 37

sentiment intensity metrics summary

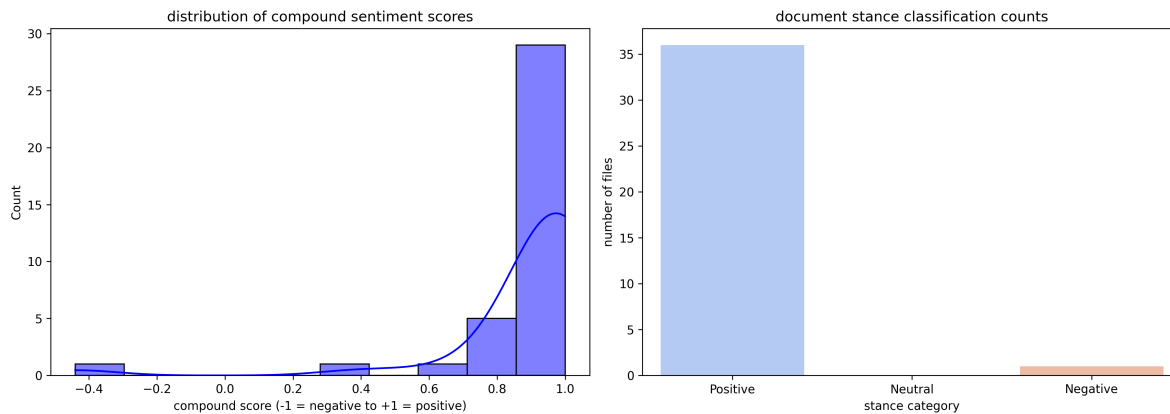
	negative score	neutral score	positive score	compound score
count	37.000000	37.000000	37.000000	37.000000
mean	0.031865	0.876459	0.091703	0.901476
std	0.020150	0.045143	0.037764	0.257008
min	0.000000	0.784000	0.047000	-0.440600
25%	0.016000	0.855000	0.062000	0.927900
50%	0.031000	0.877000	0.083000	0.997800
75%	0.043000	0.916000	0.113000	0.999800
max	0.083000	0.953000	0.185000	1.000000

categorical distributions

overall stance

Positive 36

Negative 1



Sentiment around AI tends to be positive in these txt files

US Federal laws — Sentiment Analysis

```
target_folder = "US federal laws"

sentiment_results_df = sentiment_analyzer(folder_path=target_folder)
```

sentiment analysis for folder: 'US federal laws

total text files found: 572

sentiment intensity metrics summary

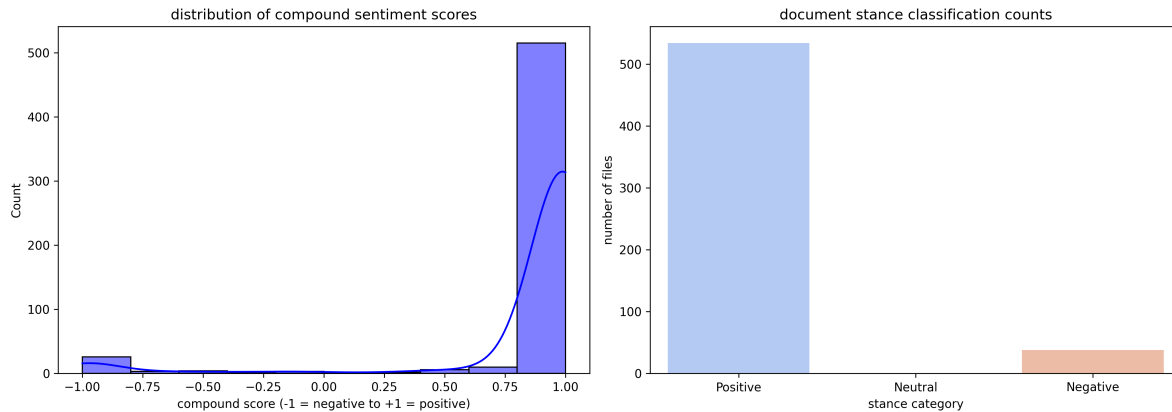
	negative score	neutral score	positive score	compound score
count	572.000000	572.000000	572.000000	572.000000
mean	0.023839	0.866886	0.109229	0.853366
std	0.026124	0.047841	0.044986	0.456189
min	0.000000	0.677000	0.010000	-0.999500
25%	0.006000	0.842750	0.076000	0.978300
50%	0.015000	0.873000	0.105000	0.996700
75%	0.035000	0.897000	0.132250	0.999200
max	0.146000	0.989000	0.288000	1.000000

categorical distributions

overall stance

Positive 534

Negative 38



Overall sentiment around AI tends to be positive in the US federal laws evaluated, but less so than other categories

US state and local documents - Sentiment Analysis

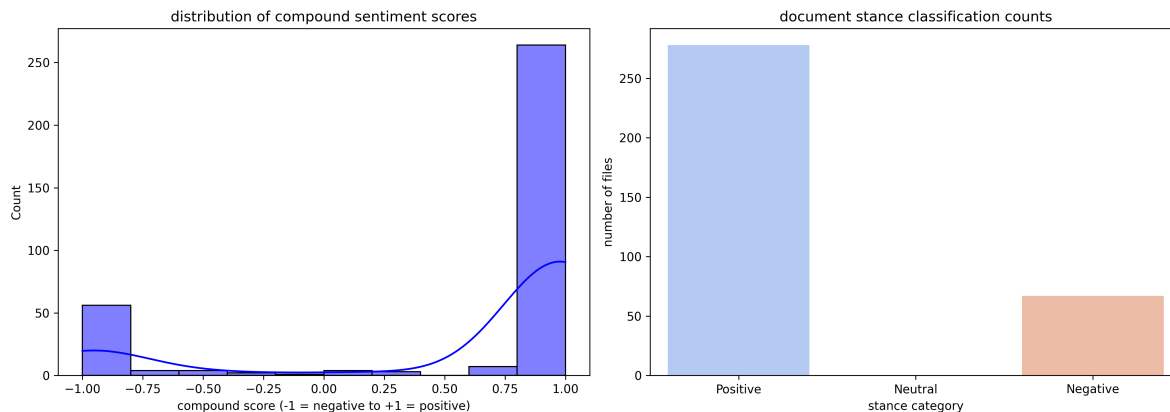
```
target_folder = "US state and local documents"

sentiment_results_df = sentiment_analyzer(folder_path=target_folder)
```

```
sentiment analysis for folder: 'US state and local documents'
total text files found: 345
sentiment intensity metrics summary
```

	negative score	neutral score	positive score	compound score
count	345.000000	345.000000	345.000000	345.000000
mean	0.039557	0.868014	0.092446	0.594285
std	0.031180	0.043203	0.041480	0.745976
min	0.000000	0.740000	0.019000	-0.999900
25%	0.017000	0.837000	0.060000	0.844400
50%	0.032000	0.872000	0.083000	0.989000
75%	0.058000	0.899000	0.116000	0.998500
max	0.153000	0.965000	0.248000	1.000000

```
categorical distributions
overall stance
Positive    278
Negative    67
```



US State and local documents appear to have the highest level of polarization when it comes to AI, although the overall stance is still positive.

US federal lawsNDAA provisionsFY2021 NDAA - Sentiment Analysis

```
target_folder = "US federal lawsNDAA provisionsFY2021 NDAA"

sentiment_results_df = sentiment_analyzer(folder_path=target_folder)
```

sentiment analysis for folder: 'US federal lawsNDAA provisionsFY2021 NDAA

total text files found: 24

sentiment intensity metrics summary

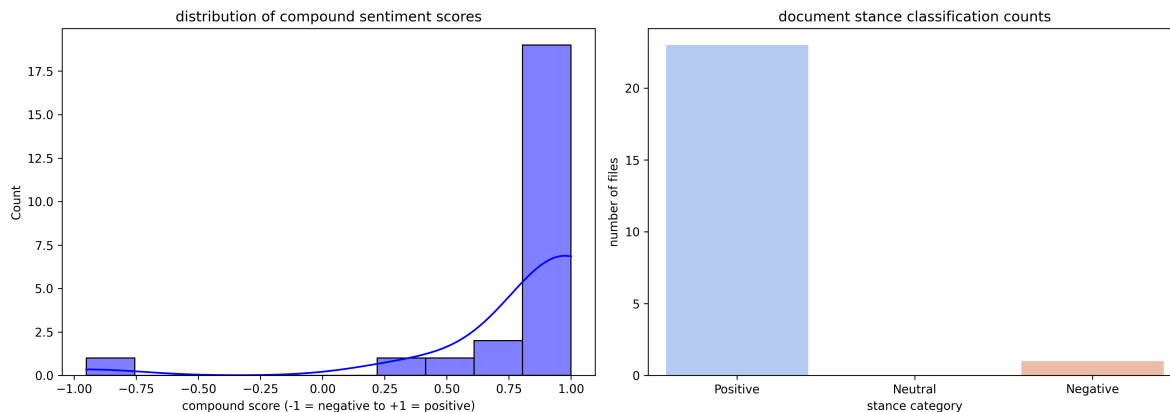
	negative score	neutral score	positive score	compound score
count	24.000000	24.000000	24.000000	24.000000
mean	0.018333	0.886583	0.094917	0.831354
std	0.019796	0.038720	0.039519	0.425318
min	0.000000	0.820000	0.024000	-0.951400
25%	0.002750	0.866250	0.079500	0.968850
50%	0.012500	0.890500	0.097500	0.991000
75%	0.026000	0.903250	0.121500	0.995400
max	0.077000	0.976000	0.160000	0.999900

categorical distributions

overall stance

Positive 23

Negative 1



The 2021 National Defense Authorization Act has mostly positive sentiment.

US federal lawsNDAA provisionsFY2022 NDAA - Sentiment Analysis

```
target_folder = "US federal lawsNDAA provisionsFY2022 NDAA"

sentiment_results_df = sentiment_analyzer(folder_path=target_folder)
```

sentiment analysis for folder: 'US federal lawsNDAA provisionsFY2022 NDAA'

total text files found: 12

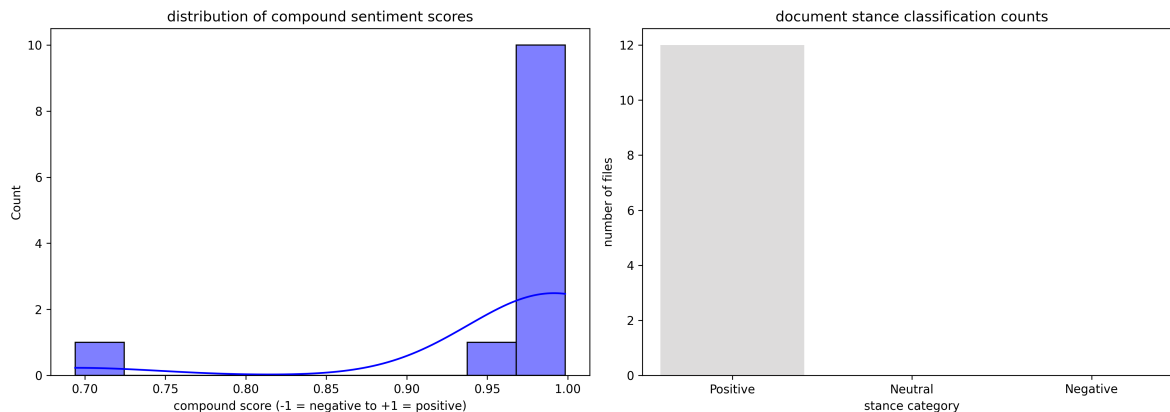
sentiment intensity metrics summary

	negative score	neutral score	positive score	compound score
count	12.000000	12.000000	12.000000	12.000000
mean	0.007833	0.869333	0.122583	0.966192
std	0.011885	0.034629	0.038349	0.086417
min	0.000000	0.799000	0.059000	0.694000
25%	0.000000	0.850500	0.103500	0.991450
50%	0.004000	0.865500	0.126000	0.992850
75%	0.009500	0.887250	0.141750	0.996200
max	0.041000	0.932000	0.201000	0.998400

categorical distributions

overall stance

Positive 12



The 2021 National Defense Authorization Act demonstrates positive sentiment based on the 12 documents rds AI

US federal lawsNDAA provisionsFY2023 NDAA - Sentiment Analysis

```
target_folder = "US federal lawsNDAA provisionsFY2023 NDAA"

sentiment_results_df = sentiment_analyzer(folder_path=target_folder)
```

sentiment analysis for folder: 'US federal lawsNDAA provisionsFY2023 NDAA

total text files found: 30

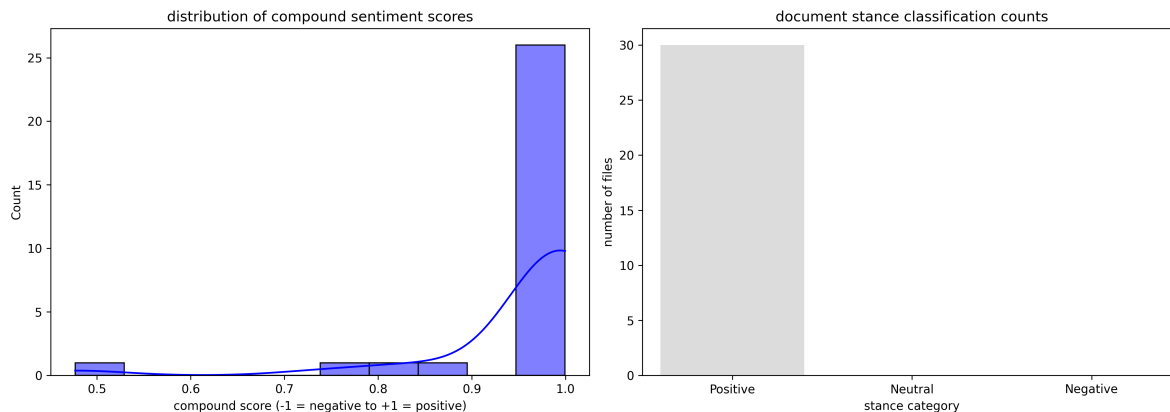
sentiment intensity metrics summary

	negative score	neutral score	positive score	compound score
count	30.000000	30.000000	30.000000	30.000000
mean	0.012033	0.861833	0.126267	0.958847
std	0.014602	0.052141	0.051811	0.107907
min	0.000000	0.750000	0.024000	0.476700
25%	0.000250	0.827250	0.086000	0.992350
50%	0.005500	0.861000	0.124000	0.996200
75%	0.019250	0.901250	0.161500	0.998125
max	0.056000	0.976000	0.235000	0.999600

categorical distributions

overall stance

Positive 30



The 2023 National Defense Authorization Act continues to show positive sentiment.

US federal lawsNDAA provisionsFY2024 NDAA - Sentiment Analysis

```
target_folder = "US federal lawsNDAA provisionsFY2024 NDAA"

sentiment_results_df = sentiment_analyzer(folder_path=target_folder)
```

sentiment analysis for folder: 'US federal lawsNDAA provisionsFY2024 NDAA'

total text files found: 22

sentiment intensity metrics summary

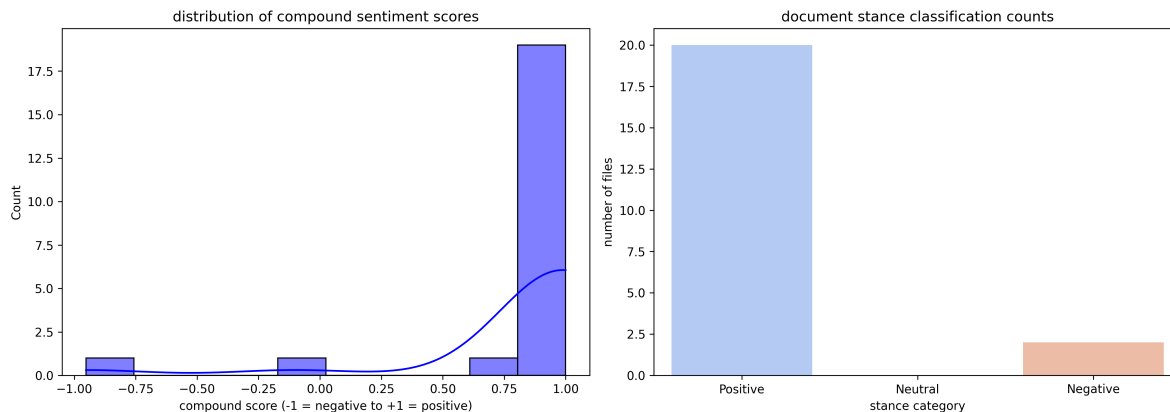
	negative score	neutral score	positive score	compound score
count	22.000000	22.000000	22.000000	22.000000
mean	0.014364	0.862045	0.123591	0.845109
std	0.028181	0.045268	0.045011	0.466722
min	0.000000	0.753000	0.054000	-0.952200
25%	0.003000	0.833250	0.090000	0.993050
50%	0.007500	0.867500	0.117500	0.995650
75%	0.009750	0.892500	0.154500	0.998250
max	0.122000	0.946000	0.241000	1.000000

categorical distributions

overall stance

Positive 20

Negative 2



The 2024 National Defense Authorization Act shows a slight uptick in potentially negative sentiment.

NDAAs provisionsUS federal lawsFY2026 NDAA- Sentiment Analysis

```
target_folder = "NDAA provisionsUS federal lawsFY2026 NDAA"

sentiment_results_df = sentiment_analyzer(folder_path=target_folder)
```

sentiment analysis for folder: 'NDAA provisionsUS federal lawsFY2026 NDAA'

total text files found: 37

sentiment intensity metrics summary

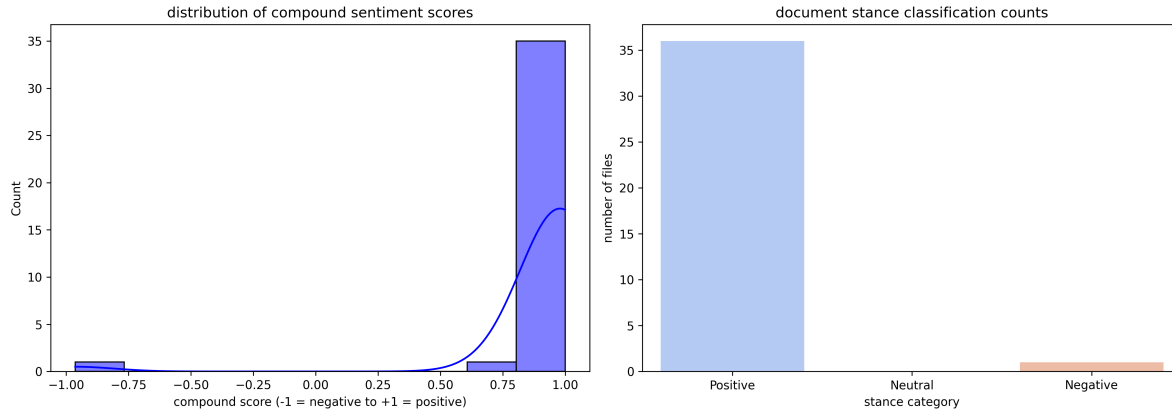
	negative score	neutral score	positive score	compound score
count	37.000000	37.000000	37.000000	37.000000
mean	0.016730	0.849595	0.133568	0.920608
std	0.027258	0.054289	0.049465	0.322611
min	0.000000	0.725000	0.067000	-0.963800
25%	0.005000	0.813000	0.092000	0.977900
50%	0.011000	0.860000	0.117000	0.993000
75%	0.018000	0.896000	0.170000	0.996700
max	0.160000	0.931000	0.254000	0.999900

categorical distributions

overall stance

Positive 36

Negative 1



The 2026 National Defense Authorization Act shows mostly positive sentiment.

Chinese law and policy - Sentiment Analysis

```
target_folder = "Chinese law and policy"
sentiment_results_df = sentiment_analyzer(folder_path=target_folder)
```

sentiment analysis for folder: 'Chinese law and policy'

total text files found: 26

sentiment intensity metrics summary

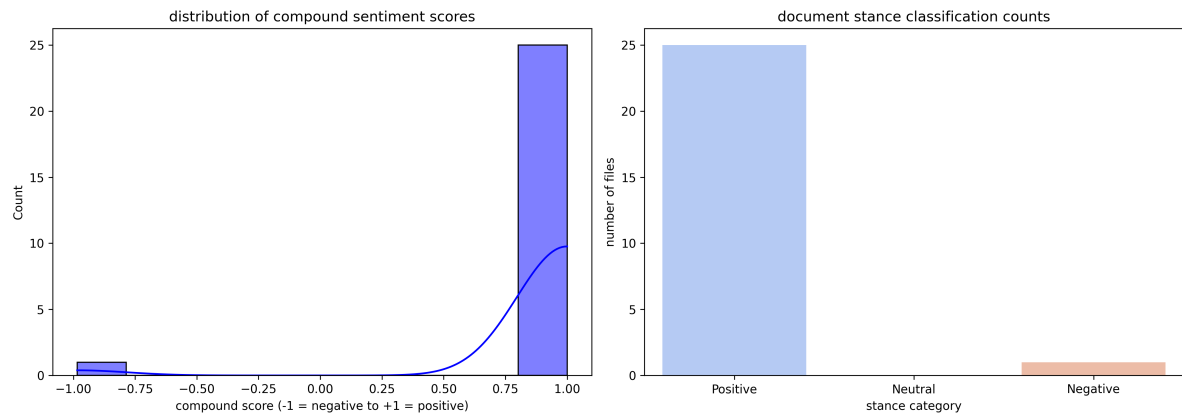
	negative score	neutral score	positive score	compound score
count	26.000000	26.000000	26.000000	26.000000
mean	0.044962	0.788462	0.166577	0.922854
std	0.028871	0.053413	0.056827	0.389069
min	0.003000	0.634000	0.082000	-0.984700
25%	0.025750	0.769500	0.122250	0.999300
50%	0.045500	0.805500	0.159500	0.999700
75%	0.058750	0.820500	0.187500	0.999875
max	0.129000	0.869000	0.299000	1.000000

categorical distributions

overall stance

Positive 25

Negative 1



Mostly postivie with 1 document with negative sentiment.